



Rex: A Family of Reversible Exponential (Stochastic) Runge-Kutta Solvers

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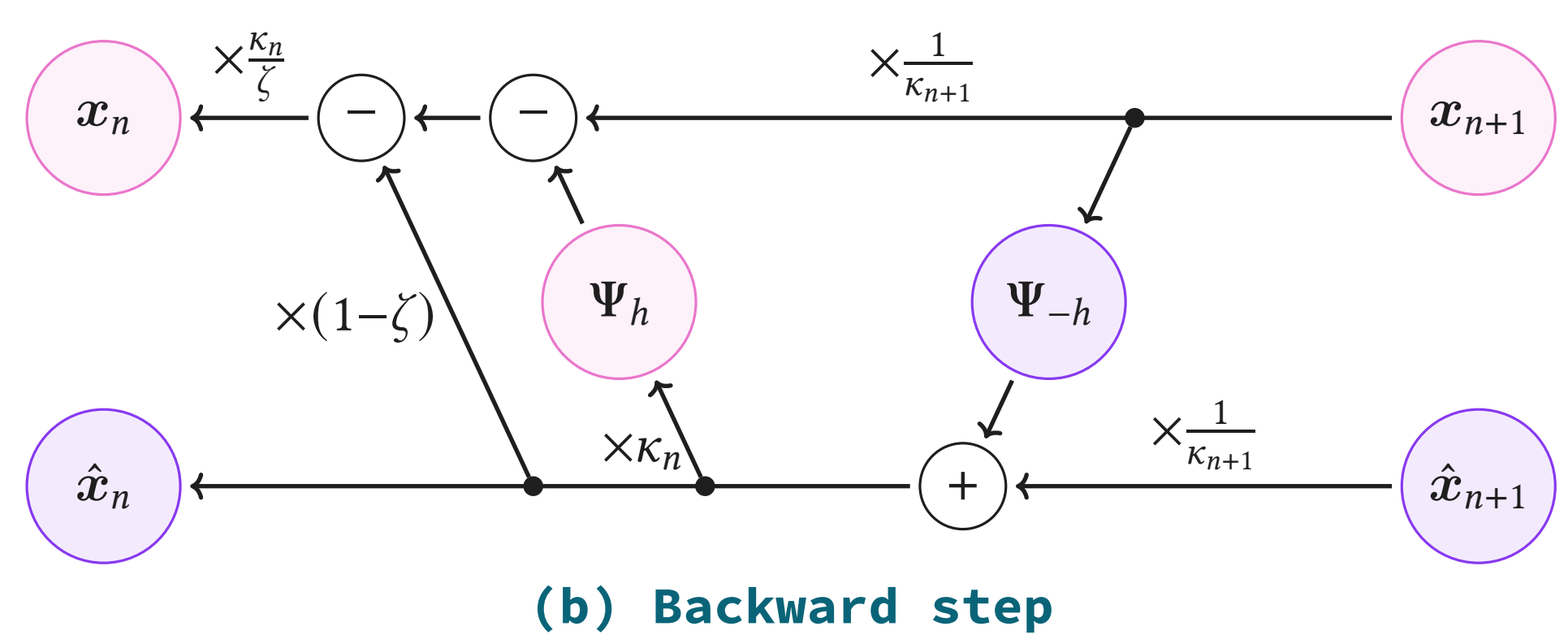
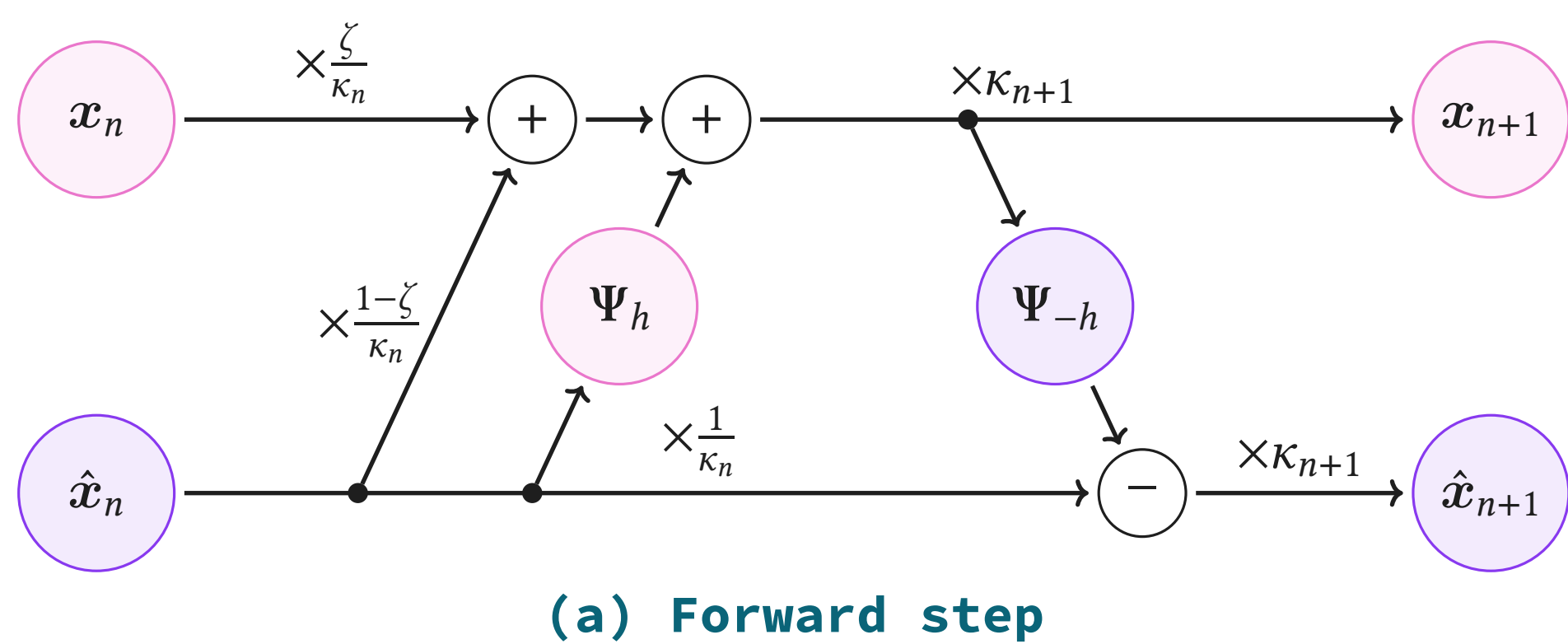


Figure 1. The Rex computation graph. Ψ_h is the exponentially weighted (S)RK scheme, $\zeta \in (0,1)$ is the coupling parameter, and κ_n is weighting term derived from the ODE/SDE formulation and if the model is parametrized via noise or data prediction. **Pink** = primary state x_n ; **mauve** = auxiliary $\hat{x}_n$. Backward step is the exact algebraic inverse.

Motivation

Diffusion models integrate between prior and data distributions via ODE/SDE solvers. Many tasks require **exact inversion**. Non-reversible solvers break this:

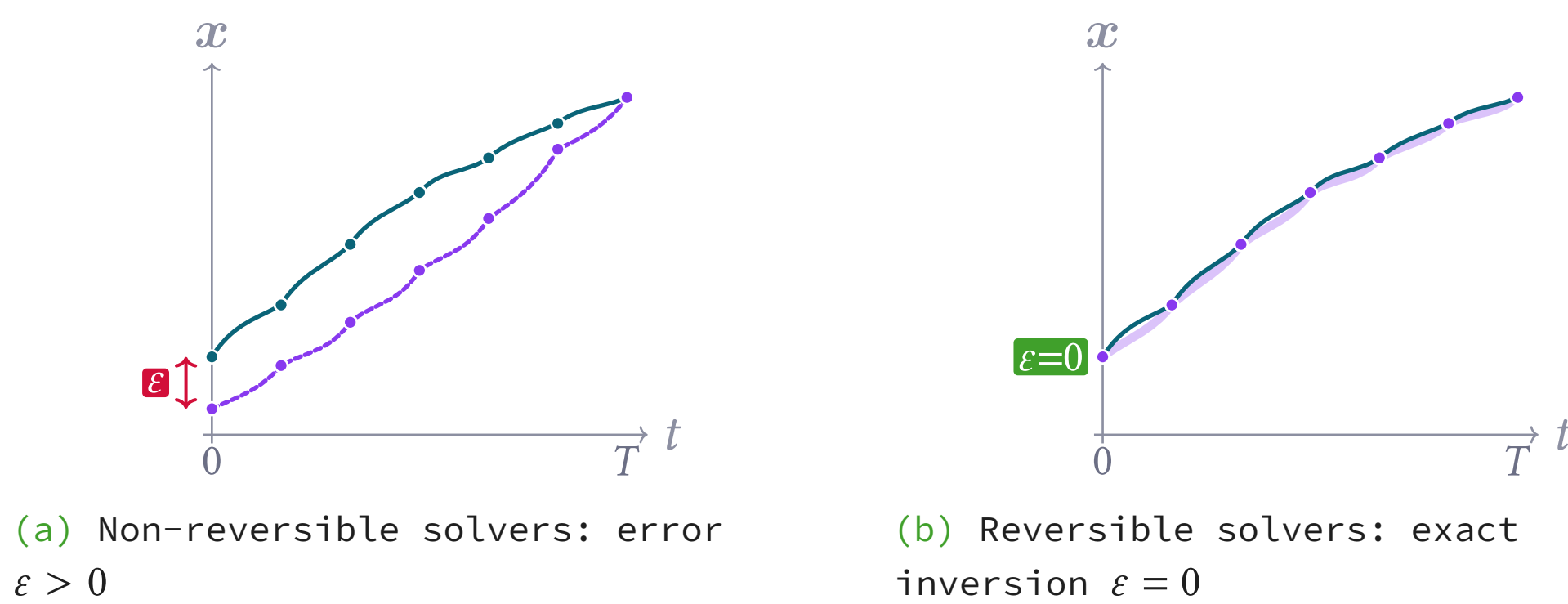


Figure 2. Teal = forward; mauve = backward. Non-reversible solvers incur reconstruction errors; algebraically reversible solvers avoid this (up to numerics).

Prior reversible solvers for diffusion models suffer from:

- **Zero/first-order** convergence (EDICT / BDIA)
- **No linear stability** (all prior methods)
- **No SDE inversion**

Exact inversion (reconstruction error)

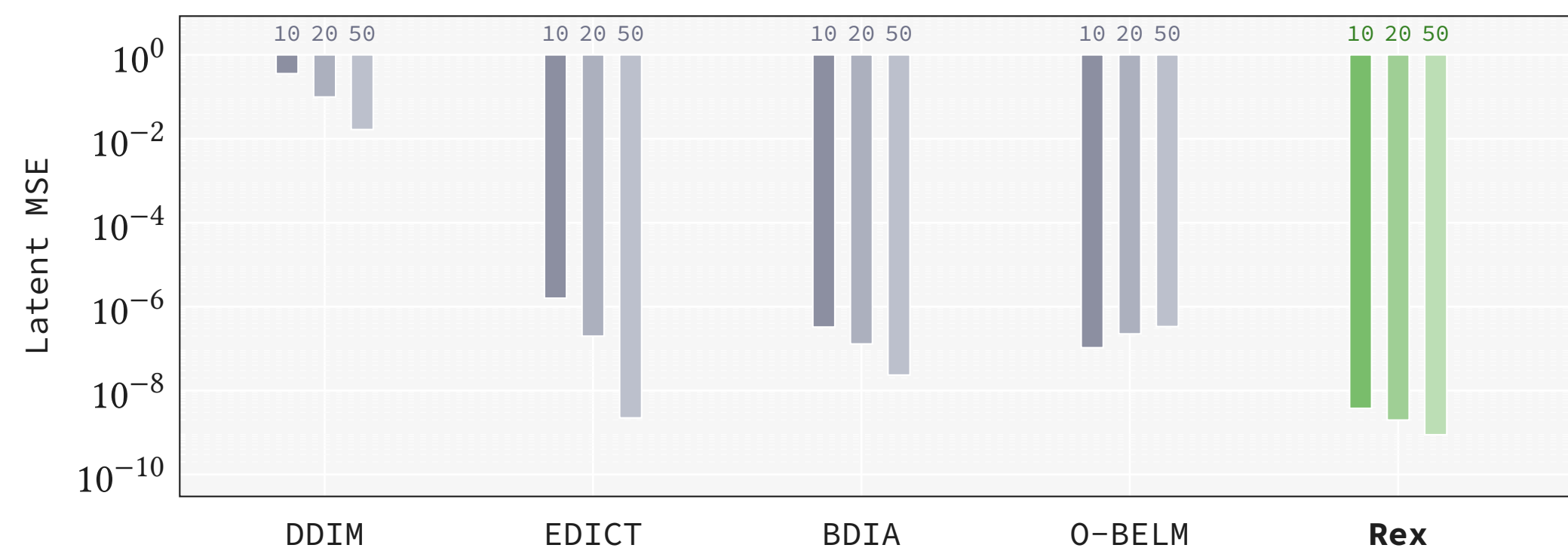


Figure 3. Numerical stability explored via the latent-space reconstruction MSE with Stable Diffusion v1.5 at fp32 precision. Numbers at top indicate discretization steps (dark = fewer). **Rex** achieves $\sim 10^{-9}$ – 10^{-10} , orders of magnitude below all baselines. Lower is better.

Rex recipe

We propose Rex a reversible family of exponential (stochastic) Runge-Kutta solvers. The recipe is as follows:

1. Select an explicit (S)RK scheme, Φ (e.g. Euler, Dopri5, ShARK).
2. Build *Princes* Ψ from Φ via exponential integrators & Lawson methods.
3. Build **Rex** out of Ψ via the McCallum-Foster framework.

Theoretical results

Theorem 1 (Convergence). Let Φ be k -th order explicit RK. Then Rex from Φ is k -th order: $\|x_n - x_{t_n}\| \leq Ch^k$.

Theorem 2 (Subsumes prior solvers). Rex is the reversible version of DDIM, DPM-Solver-1/2/12, DPM-Solver++1/(2S), SDE-DPM-Solver-1/++1, SEEDS-1, and gDDIM.

Image generation & editing



Figure 4. Text-to-image with SD v1.5, 10 steps.

Table 1. Quantitative comparison of different reversible solvers image editing with the non-reversible DDIM as a baseline. Our reversible ODE solvers are in **pink** and adaptive ODE solvers are in **teal**. Best results in **bold**, second best under-lined.

Solver	Image Reward ($\uparrow$)	CLIP score ($\uparrow$)	PickScore ($\uparrow$)	LPIPS ($\downarrow$)
DDIM	-0.564	19.17	18.367	0.214
BDIA	-2.205	18.57	16.956	0.885
O-BELM	-0.639	19.16	18.416	0.140
Rex (Euler)	<u>-0.551</u>	19.17	18.721	<u>0.109</u>
Rex (Dopri5)	-0.547	<u>19.16</u>	<u>18.698</u>	0.107

Boltzmann generators

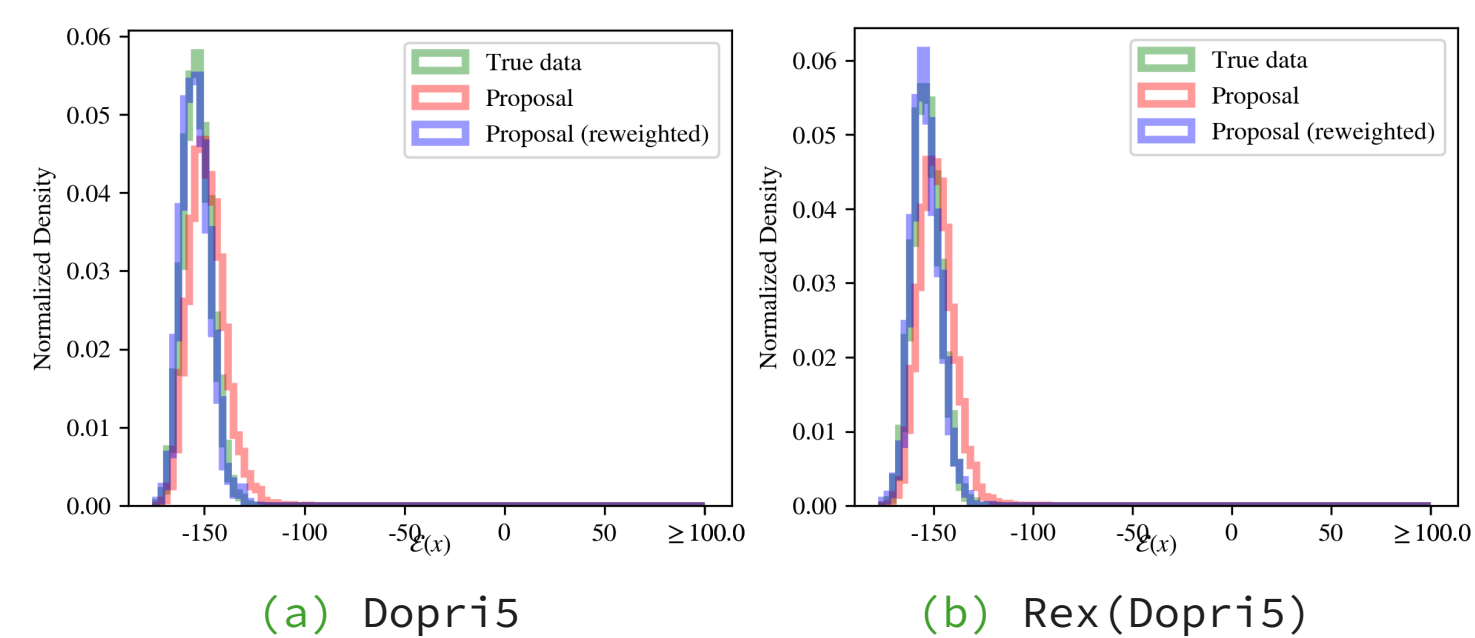


Figure 5. True MD energy distribution with unweighted and reweighted proposals created using Rex(Dopri5).

Table 2. Quantitative results on tri-alanine. Evaluations are conducted over 10^4 samples. Best results in **bold**, second best under-lined.

Model	Numerical scheme	ESS ($\uparrow$)	$\mathcal{E}\text{-}\mathcal{W}_2(\downarrow)$	$\mathcal{T}\text{-}\mathcal{W}_2(\downarrow)$
RegFlow	–	0.029	1.051	1.612
SBG (IS)	–	0.052	0.758	0.502
SBG (SMC)	–	–	0.598	0.503
ECNF++	Dopri5	0.003	2.206	0.962
DiT	Dopri5	0.140	<u>0.737</u>	0.468
DiT	Rex (Dopri5)	<u>0.104</u>	0.495	<u>0.497</u>

Key takeaways

1. **Exact algebraic inversion** for both diffusion ODEs and SDEs.
2. Inherits **convergence order** of underlying RK with **non-trivial linear stability**.
3. **Rex outperforms all baselines** in terms of inversion, Image Reward, PickScore, and $\mathcal{E}\text{-}\mathcal{W}_2$.